

Artificial Intelligence: the global landscape of ethics guidelines

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Preprint version

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ABSTRACT

In the last five years, private companies, research institutions as well as public sector organisations have issued principles and guidelines for ethical AI, yet there is debate about both what constitutes “ethical AI” and which ethical requirements, technical standards and best practices are needed for its realization. To investigate whether a global agreement on these questions is emerging, we mapped and analyzed the current corpus of principles and guidelines on ethical AI. Our results reveal a global convergence emerging around five ethical principles (transparency, justice and fairness, non-maleficence, responsibility and privacy), with substantive divergence in relation to how these principles are interpreted; why they are deemed important; what issue, domain or actors they pertain to; and how they should be implemented. Our findings highlight the importance of integrating guideline-development efforts with substantive ethical analysis and adequate implementation strategies.

MAIN ARTICLE

Introduction

Artificial Intelligence (AI), or the theory and development of computer systems able to perform tasks normally requiring human intelligence, is widely heralded as an ongoing “revolution” transforming science and society altogether^{1,2}. While approaches to AI such as machine learning, deep learning and artificial neural networks are reshaping data processing and analysis³, autonomous and semi-autonomous systems are being increasingly used in a variety of sectors including healthcare, transportation and the production chain⁴. In light of its powerful transformative force and profound impact across various societal domains, AI has sparked ample debate about the principles and values that should guide its development and use^{5,6}. Fears that AI might jeopardize jobs for human workers⁷, be misused by malevolent actors⁸, elude accountability or inadvertently disseminate bias and thereby undermine fairness⁹ have been at the forefront of the recent scientific literature and media coverage. Several studies have discussed the topic of ethical AI^{10–13}, notably in meta-assessments^{14–16} or in relation to systemic risks^{17,18} and unintended negative consequences like algorithmic bias or discrimination^{19–21}.

National and international organisations have responded to these societal fears by developing *ad hoc* expert committees on AI, often commissioned with the drafting of policy documents. These include the High-Level Expert Group on Artificial Intelligence appointed by the European Commission, the expert group on AI in Society of the Organisation for Economic Co-operation and Development (OECD), the Advisory Council on the Ethical Use of Artificial Intelligence and Data in Singapore, and the select committee on Artificial Intelligence of the United Kingdom (UK) House of Lords. As part of their institutional appointments, these committees have produced or are reportedly producing reports and guidance documents on AI. Similar efforts are taking place in the private sector, especially among corporations who rely on AI for their business. In 2018 alone, companies such as Google and SAP have publicly released AI guidelines and principles. Declarations and recommendations have also been issued by professional associations and non-profit organisations such as the Association of Computing Machinery (ACM), Access Now and Amnesty International. The intense efforts of such a diverse set of stakeholders in issuing AI principles and policies demonstrate not only the need for ethical guidance, but also the

strong interest of these stakeholders to shape the ethics of AI in ways that meet their respective priorities¹⁶. Notably, the private sector's involvement in the AI-ethics arena has been called into question for potentially using such high-level soft-policy as a portmanteau to either render a social problem technical¹⁶ or to eschew regulation altogether²². Beyond the composition of the groups that have produced ethical guidance on AI, the content of this guidance itself is of interest. Are these various groups converging on what ethical AI should be, and the ethical principles that will determine the development of AI? If they diverge, what are these differences and can they be reconciled?

To answer these questions, we conducted a scoping review of the existing corpus of guidelines on ethical AI. Our analysis aims at mapping the global landscape of existing principles and guidelines for ethical AI and thereby determining whether a global convergence is emerging regarding both the principles for ethical AI and the requirements for its realization. This analysis will inform scientists, research institutions, funding agencies, governmental and inter-governmental organisations and other relevant stakeholders involved in the advancement of ethically responsible innovation in AI.

Results

Our search identified 84 documents containing ethical principles or guidelines for AI (cf. Table 1). Data reveal a significant increase over time in the number of publications, with 88% having been released after 2016 (cf. SI Table S1). Data breakdown by type and geographic location of issuing organisation (cf. SI Table S1) shows that most documents were produced by private companies (n=19; 22.6%) and governmental agencies respectively (n=18; 21.4%), followed by academic and research institutions (n=9; 10.7%), inter-governmental or supra-national organisations (n=8; 9.5%), non-profit organisations and professional associations/scientific societies (n=7 each; 8.3% each), private sector alliances (n=4; 4.8%), research alliances (n=1; 1.2%), science foundations (n=1; 1.2%), federations of worker unions (n=1; 1.2%) and political parties (n=1; 1.2%). Four documents were issued by initiatives belonging to more than one of the above categories and four more could not be classified at all (4.8% each).

Table 1- Ethical guidelines for AI by country of issuer

Name of Document/Website	Issuer	Country of issuer
Artificial Intelligence. Australia's Ethics Framework. A discussion Paper	Department of Industry Innovation and Science	Australia
Montréal Declaration: Responsible AI	Université de Montréal	Canada
Work in the age of artificial intelligence. Four perspectives on the economy, employment, skills and ethics	Ministry of Economic Affairs and Employment	Finland
Tieto's AI ethics guidelines	Tieto	Finland
Commitments and principles	OP Group	Finland
How can humans keep the upper hand? Report on the ethical matters raised by AI algorithms	French Data Protection Authority (CNIL)	France
For a meaningful Artificial Intelligence. Towards a French and European strategy	Mission Villani	France
Ethique de la recherche en robotique	CERNA (Allistene)	France
AI Guidelines	Deutsche Telekom	Germany
SAP's guiding principles for artificial intelligence	SAP	Germany
Automated and Connected Driving: Report	Federal Ministry of Transport and Digital Infrastructure, Ethics Commission	Germany
Ethics Policy	Icelandic Institute for Intelligent Machines (IIIM)	Iceland
Discussion Paper: National Strategy for Artificial Intelligence	National Institution for Transforming India (Niti Aayog)	India
L'intelligenza artificiale al servizio del cittadino	Agenzia per l'Italia Digitale (AGID)	Italy
The Japanese Society for Artificial Intelligence Ethical Guidelines	Japanese Society for Artificial Intelligence	Japan
Report on Artificial Intelligence and Human Society (Unofficial translation)	Advisory Board on Artificial Intelligence and Human Society (initiative of the Minister of State for Science and Technology Policy)	Japan
Draft AI R&D Guidelines for International Discussions	Institute for Information and Communications Policy (IICP), The Conference toward AI Network Society	Japan
Sony Group AI Ethics Guidelines	SONY	Japan
Human Rights in the Robot Age Report	The Rathenau Institute	Netherlands
Dutch Artificial Intelligence Manifesto	Special Interest Group on Artificial Intelligence (SIGAI), ICT Platform Netherlands (IPN)	Netherlands
Artificial intelligence and privacy	The Norwegian Data Protection Authority	Norway
Discussion Paper on Artificial Intelligence (AI) and Personal Data - Fostering Responsible Development and Adoption of AI	Personal Data Protection Commission Singapore	Singapore
Mid- to Long-Term Master Plan in Preparation for the Intelligent Information Society	Government of the Republic of Korea	South Korea
AI Principles of Telefónica	Telefonica	Spain
AI Principles & Ethics	Smart Dubai	UAE
Principles of robotics	Engineering and Physical Sciences Research Council UK (EPSRC)	UK
The Ethics of Code: Developing AI for Business with Five Core Principles	Sage	UK
Big data, artificial intelligence, machine learning and data protection	Information Commissioner's Office	UK
DeepMind Ethics & Society Principles	DeepMind Ethics & Society	UK
Business Ethics and Artificial Intelligence	Institute of Business Ethics	UK
AI in the UK: ready, willing and able?	UK House of Lords, Select Committee on Artificial Intelligence	UK
Artificial Intelligence (AI) in Health	Royal College of Physicians	UK
Initial code of conduct for data-driven health and care technology	UK Department of Health & Social Care	UK
Ethics Framework - Responsible AI	Machine Intelligence Garage Ethics Committee	UK
The responsible AI framework	PriceWaterhouseCoopers UK	UK
Responsible AI and robotics. An ethical framework.	Accenture UK	UK
Machine learning: the power and promise of computers that learn by example	The Royal Society	UK
Ethical, social, and political challenges of Artificial Intelligence in Health	Future Advocacy	UK
Unified Ethical Frame for Big Data Analysis. IAF Big Data Ethics Initiative, Part A	The Information Accountability Foundation	UK
The AI Now Report. The Social and Economic Implications of Artificial Intelligence Technologies in the Near-Term	AI Now Institute	USA
Statement on Algorithmic Transparency and Accountability	Association for Computing Machinery (ACM)	USA
AI Principles	Future of Life Institute	USA
AI - Our approach	Microsoft	USA
Artificial Intelligence. The Public Policy Opportunity	Intel Corporation	USA
IBM's Principles for Trust and Transparency	IBM	USA
OpenAI Charter	OpenAI	USA
Our principles	Google	USA
Policy Recommendations on Augmented Intelligence in Health Care H-480 940	American Medical Association (AMA)	USA
Everyday Ethics for Artificial Intelligence. A practical guide for designers & developers	IBM	USA
Governing Artificial Intelligence. Upholding Human Rights & Dignity	Data & Society	USA
Intel's AI Privacy Policy White Paper. Protecting individuals' privacy and data in the artificial intelligence world	Intel Corporation	USA
Introducing Unity's Guiding Principles for Ethical AI – Unity Blog	Unity Technologies	USA
Digital Decisions	Center for Democracy & Technology	USA
Science, Law and Society (SLS) Initiative	The Future Society	USA
AI Now 2018 Report	AI Now Institute	USA
Responsible bots: 10 guidelines for developers of conversational AI	Microsoft	USA
Preparing for the future of Artificial Intelligence	Executive Office of the President; National Science and Technology Council; Committee on Technology	USA
The National Artificial Intelligence Research and Development Strategic Plan	National Science and Technology Council; Networking and Information Technology Research and Development Subcommittee	USA

AI Now 2017 Report	AI Now Institute	USA
Position on Robotics and Artificial Intelligence	The Greens (Green Working Group Robots)	EU
Report with recommendations to the Commission on Civil Law Rules on Robotics	European Parliament	EU
Ethics Guidelines for Trustworthy AI	High-Level Expert Group on Artificial Intelligence	EU
AI4People—An Ethical Framework for a Good AI Society: Opportunities, Risks, Principles, and Recommendations	AI4People	EU
European ethical Charter on the use of Artificial Intelligence in judicial systems and their environment	Concil of Europe: European Commission for the efficiency of Justice (CEPEJ)	EU
Statement on Artificial Intelligence, Robotics and 'Autonomous' Systems	European Commission, European Group on Ethics in Science and New Technologies	EU
Artificial Intelligence and Machine Learning: Policy Paper	Internet Society	international
Report of COMEST on Robotics Ethics	COMEST/UNESCO	international
Ethical Principles for Artificial Intelligence and Data Analytics	Software & Information Industry Association (SIIA), Public Policy Division	international
ITI AI Policy Principles	Information Technology Industry Council (ITI)	international
Ethically Aligned Design. A Vision for Prioritizing Human Well-being with Autonomous and Intelligent Systems, version 2	Institute of Electrical and Electronics Engineers (IEEE), The IEEE Global Initiative on Ethics of Autonomous and Intelligent Systems	international
Top 10 Principles for Ethical Artificial Intelligence	UNI Global Union	international
The Malicious Use of Artificial Intelligence: Forecasting, Prevention, and Mitigation	Future of Humanity Institute; University of Oxford; Centre for the Study of Existential Risk; University of Cambridge; Center for a New American Security; Electronic Frontier Foundation; OpenAI	international
White Paper: How to Prevent Discriminatory Outcomes in Machine Learning	WEF, Global Future Council on Human Rights 2016-2018	international
Privacy and Freedom of Expression In the Age of Artificial Intelligence	Privacy International & Article 19	international
The Toronto Declaration: Protecting the right to equality and non-discrimination in machine learning systems	Access Now ; Amnesty International	international
Charlevoix Common Vision for the Future of Artificial Intelligence	Leaders of the G7	international
Artificial Intelligence: open questions about gender inclusion	W20	international
Declaration on ethics and data protection in Artificial Intelligence	ICDPPC	international
Universal Guidelines for Artificial Intelligence	The Public Voice	international
Ethics of AI in Radiology: European and North American Multisociety Statement	American College of Radiology; European Society of Radiology; Radiology Society of North America; Society for Imaging Informatics in Medicine; European Society of Medical Imaging Informatics; Canadian Association of Radiologists; American Association of Physicists in Medicine	international
Ethically Aligned Design: A Vision for Prioritizing Human Well-being with Autonomous and Intelligent Systems, First Edition (EAD1e)	Institute of Electrical and Electronics Engineers (IEEE), The IEEE Global Initiative on Ethics of Autonomous and Intelligent Systems	international
Tenets	Partnership on AI	n.a.
Principles for Accountable Algorithms and a Social Impact Statement for Algorithms	Fairness, Accountability, and Transparency in Machine Learning (FATML)	n.a.
10 Principles of responsible AI	Women leading in AI	n.a.

In terms of geographic distribution, data show a significant representation of more economically developed countries (MEDC), with the USA (n=20; 23.8%) and the UK (n=14; 16.7%) together accounting for more than a third of all ethical AI principles, followed by Japan (n=4; 4.8%), Germany, France, and Finland (each n=3; 3.6% each). The cumulative number of sources from the European Union, comprising both documents issued by EU institutions (n=6) and documents issued within each member state (13 in total), accounts for 19 documents overall. African and South-American countries are not represented independently from international or supra-national organisations (cf. Figure 1).

Figure 1- Geographic distribution of issuers of ethical AI guidelines by number of documents released

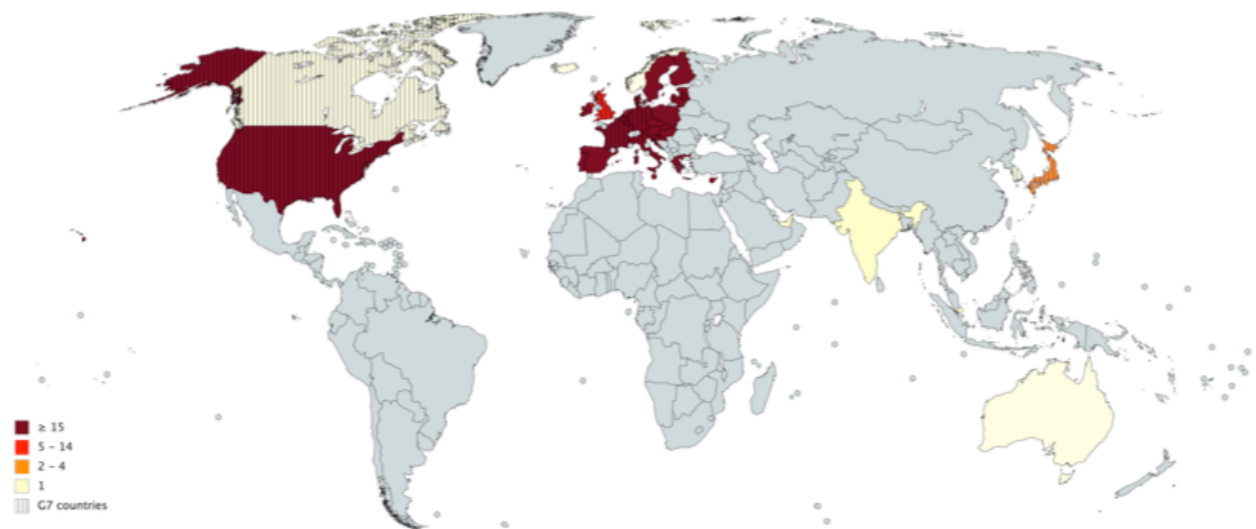


Figure 1: Geographic distribution of issuers of ethical AI guidelines by number of documents released. Most ethics guidelines are released in the United States (n=20) and within the European Union (19), followed by the United Kingdom (14) and Japan (4). Canada, Iceland, Norway, the United Arab Emirates, India, Singapore, South Korea, Australia are represented with 1 document each. Having endorsed a distinct G7 statement, member states of the G7 countries are highlighted separately. Map created using mapchart.net.

Data breakdown by target audience indicates that most principles and guidelines are addressed to multiple stakeholder groups (n=27; 32.1%). Another significant portion of the documents is self-directed, as they are addressed to a category of stakeholders within the sphere of activity of the issuer such as the members of the issuing organisation or the issuing company's employees (n=24; 28.6%). Finally, some documents target the public sector (n=10; 11.9%), the private sector (n=5; 6.0%), or other specific stakeholders beyond members of the issuing organisation, namely developers or designers (n=3; 3.6%), 'organisations' (n=1; 1.2%) and researchers (n=1; 1.2%). 13 sources (15.5%) do not specify their target audience (cf. SI Table S1).

Eleven overarching ethical values and principles have emerged from our content analysis. These are, by frequency of the number of sources in which they were featured: transparency, justice and fairness, non-maleficence, responsibility, privacy, beneficence, freedom and autonomy, trust, dignity, sustainability, and solidarity (cf. Table 2).

Table 2 – Ethical principles identified in existing AI guidelines

Ethical principle	Number of documents	Included codes
Transparency	73/84	Transparency, explainability, explicability, understandability, interpretability, communication, disclosure, showing
Justice & fairness	68/84	Justice, fairness, consistency, inclusion, equality, equity, (non-)bias, (non-)discrimination, diversity, plurality, accessibility, reversibility, remedy, redress, challenge, access and distribution
Non-maleficence	60/84	Non-maleficence, security, safety, harm, protection, precaution, prevention, integrity (bodily or mental), non-subversion
Responsibility	60/84	Responsibility, accountability, liability, acting with integrity
Privacy	47/84	Privacy, personal or private information
Beneficence	41/84	Benefits, beneficence, well-being, peace, social good, common good
Freedom & autonomy	34/84	Freedom, autonomy, consent, choice, self-determination, liberty, empowerment
Trust	28/84	Trust
Sustainability	14/84	Sustainability, environment (nature), energy, resources (energy)
Dignity	13/84	Dignity
Solidarity	6/84	Solidarity, social security, cohesion

No single ethical principle appeared to be common to the entire corpus of documents, although there is an emerging convergence around the following principles: transparency, justice and fairness, non-maleficence, responsibility, and privacy. These principles are referenced in more than half of all the sources. Nonetheless, further thematic analysis reveals significant semantic and conceptual divergences in both how the eleven ethical principles are interpreted and the specific recommendations or areas of concern derived from each. A detailed thematic evaluation is presented in the following.

Transparency

Featured in 73/84 sources, transparency is the most prevalent principle in the current literature. Thematic analysis reveals significant variation in relation to the interpretation,

justification, domain of application, and mode of achievement. References to transparency comprise efforts to increase explainability, interpretability or other acts of communication and disclosure (cf. Table 2). Principal domains of application include data use^{23–26}, human-AI interaction^{23,27–35}, automated decisions^{26,36–46}, and the purpose of data use or application of AI systems^{24,27,47–51}. Primarily, transparency is presented as a way to minimize harm and improve AI^{36–38,44,45,49,52–55}, though some sources underline its benefit for legal reasons^{37,45,46,49,50,52} or to foster trust^{23,24,29,33,36,37,48,51,52,56–58}. A few sources also link transparency to dialogue, participation, and the principles of democracy^{30,41,49,50,52,59}.

To achieve greater transparency, many sources suggest increased disclosure of information by those developing or deploying AI systems^{36,51,60,61}, although specifications regarding *what* should be communicated vary greatly: use of AI⁴⁵, source code^{31,52,62}, data use^{35,47,50,58}, evidence base for AI use⁵⁷, limitations^{25,33,47,51,58,60,63}, laws^{62,64}, responsibility for AI⁴⁰, investments in AI^{44,65} and possible impact⁶⁶. The provision of explanations ‘in non-technical terms’²⁶ or auditable by humans^{37,60} is encouraged. Whereas audits and auditability^{28,39,44,45,50,59,61,62,67,68} are mainly proposed by data protection offices and NPOs, it is mostly the private sector that suggests technical solutions^{27,30,52,59,69,70}. Alternative measures focus on oversight^{45,47,48,55,62}, interaction and mediation with stakeholders and the public^{24,32,36,51,61,71} and the facilitation of whistleblowing^{36,60}.

Justice, fairness, and equity

Justice is mainly expressed in terms of fairness^{23,25,27–29,48,50,58,60,66,72–77}, and of prevention, monitoring or mitigation of unwanted bias^{23,28,33,40,47,52,54,58,64,69,73,74,78–80} and discrimination^{28,33,36,38,44,45,50,55,56,60,68,81–84}, the latter being significantly less referenced than the first two by the private sector. Whereas some sources focus on justice as respect for diversity^{31,38,56,59,65,66,70,72,78,80,85,86}, inclusion^{31,45,47,51,72,80} and equality^{41,45,51,59,60,72,78}, others call for a possibility to appeal or challenge decisions^{28,35–37,74,79}, or the right to redress^{33,42,45,46,50,68,85} and remedy^{45,48}. Sources also emphasize the importance of fair access to AI^{59,70,87}, to data^{33,37,44,67,83,88–90}, and to the benefits of AI^{37,38,80,91}. Issuers from the public sector place particular emphasis on AI’s impact on the labor market^{37,38,55,84,92}, and the need to address democratic^{33,38,59,73} or societal^{31,48,55,65} issues. Sources focusing on the risk of biases within datasets underline the importance of acquiring and processing accurate, complete and diverse data^{23,28,52,70,93}, especially training data^{27,33,35,38,52,58}.

If specified, the preservation and promotion of justice are proposed to be pursued through: (a) technical solutions such as standards^{50,68,89} or explicit normative encoding^{28,37,43,67}; (b) transparency^{54,62}, notably by providing information^{36,38,79} and raising public awareness of existing rights and regulation^{28,59}; (c) testing^{52,58,67,69}, monitoring^{54,56} and auditing^{39,46,50,67}, the preferred solution of notably data protection offices; (d) developing or strengthening the rule of law and the right to appeal, recourse, redress, or remedy^{37,38,42,45,46,48,68,74,79}; (e) via systemic changes and processes such as governmental action^{42,45,87,92} and oversight⁹⁴, a more interdisciplinary^{47,65,85,93} or otherwise diverse^{58,59,70,85,87,95} workforce, as well as better inclusion of civil society or other relevant stakeholders in an interactive manner^{28,33,41,46,55,57,58,65,68,69,79,80,86} and increased attention to the distribution of benefits^{25,33,38,48,63,76}.

Non-maleficence

References to non-maleficence outweigh those to beneficence by a factor of 1.5 and encompass general calls for safety and security^{80,90,96,97} or state that AI should never cause foreseeable or unintentional harm^{23,30,33,56,60,79}. More granular considerations entail the avoidance of specific risks or potential harms, e.g. intentional misuse via cyberwarfare and malicious hacking^{51,53,54,78,81,89}, and suggest risk-management strategies. Harm is primarily interpreted as discrimination^{38,44,47,48,50,95,98}, violation of privacy^{23,35,44,64,78,98,99}, or bodily harm^{25,30,31,33,56,92,96,100}. Less frequent characterizations include loss of trust³⁰ or skills⁴⁴, ‘radical individualism’³⁸, the risk that technological progress might outpace regulatory measures⁵⁷, negative impacts on long-term social well-being⁴⁴, on infrastructure⁴⁴, or on psychological^{35,56}, emotional⁵⁶ or economic aspects^{44,56}.

Harm-prevention guidelines focus primarily on technical measures and governance strategies, ranging from interventions at the level of AI research^{27,47,64,79,85,101}, design^{23,25,27,32,39,56,58}, technology development and/or deployment⁵⁴ to lateral and continuous approaches^{33,55,63}. Technical solutions include in-built data quality evaluations²⁵ or security²³ and privacy by design^{23,27,39}, though notable exceptions also advocate for establishing industry standards^{30,64,102}. Proposed governance strategies include active cooperation across disciplines and stakeholders^{33,47,53,62}, compliance with existing or new legislation^{27,31,35,81,95,99}, and the need to establish oversight processes and practices, notably

tests^{36,38,47,74,79}, monitoring^{36,58}, audits and assessments by internal units, customers, users, independent third parties, or governmental entities^{40,48,51,58,81,94,95,98}, often geared towards standards for AI implementation and outcome assessment. Most sources explicitly mention potential ‘dual-use’^{8,32,33,38,60,79} or imply that damages may be unavoidable, in which case risks should be assessed^{40,48,51}, reduced^{40,69,72–74}, and mitigated^{34,35,38,53,63,68}, and the attribution of liability should be clearly defined^{31,37,38,44,82}.

Responsibility and accountability

Despite widespread references to ‘responsible AI’^{43,51,78,83}, responsibility and accountability are rarely defined. Nonetheless, specific recommendations include acting with ‘integrity’^{47,52,60} and clarifying the attribution of responsibility and legal liability^{23,58,78,103}, if possible upfront³⁶, in contracts⁵² or, alternatively, by centering on remedy²⁶. In contrast, other sources suggest focusing on the underlying reasons and processes that may lead to potential harm^{74,83}. Yet others underline the responsibility of whistleblowing in case of potential harm^{36,55,60}, and aim at promoting diversity^{49,92} or introducing ethics into STEM education⁵⁹. Very different actors are named as being responsible and accountable for AI’s actions and decisions: AI developers^{58,60,73,96}, designers^{36,44}, ‘institutions’^{40,42} or ‘industry’⁶⁹. Further disagreement emerged on whether AI should be held accountable in a human-like manner⁷⁰ or whether humans should always be the only actors who are ultimately responsible for technological artifacts^{31,32,35,37,52,92}.

Privacy

Ethical AI sees privacy both as a value to uphold^{44,64,75,99} and as a right to be protected^{27,28,37,38,53}. While often undefined, privacy is often presented in relation to data protection^{23,27,36,53,58,66,71,79,83,98} and data security^{27,35,64,66,88,98}. A few sources link privacy to freedom^{38,53} or trust^{74,92}. Suggested modes of achievement fall into three categories: technical solutions^{64,80} such as differential privacy^{74,89}, privacy by design^{25,27,28,79,98}, data minimization^{36,58}, and access control^{36,58}, calls for more research^{47,64,74,98} and awareness^{64,74}, and regulatory approaches^{25,52,71}, with sources referring to legal compliance more broadly^{27,32,36,58,60,81}, or suggesting certificates¹⁰⁴ or the creation or adaptation of laws and regulations to accommodate the specificities of AI^{64,74,88,105}.

Beneficence

While promoting good (*beneficence* in ethical terms) is often mentioned, it is rarely defined, though notable exceptions mention the augmentation of human senses⁸⁶, the promotion of human well-being and flourishing^{34,90}, peace and happiness⁶⁰, the creation of socio-economic opportunities³⁶, and economic prosperity^{37,53}. Similar uncertainty concerns the actors that should benefit from AI: private sector issuers tend to highlight the benefit of AI for customers^{23,48}, though many sources require AI to be shared^{49,52,76} and to benefit ‘everyone’^{36,59,65,84}, ‘humanity’^{27,37,44,60,100,102}, both of the above^{48,66}, ‘society’^{34,87}, ‘as many people as possible’^{37,53,99}, ‘all sentient creatures’⁸³, the ‘planet’^{37,72} and the environment^{38,90}. Strategies for the promotion of good include aligning AI with human values^{34,44}, advancing ‘scientific understanding of the world’¹⁰⁰, minimizing power concentration¹⁰² or, conversely, using power ‘for the benefit of human rights’⁸²; working more closely with ‘affected’ people⁶⁵, minimizing conflicts of interests¹⁰²; proving beneficence through customer demand⁴⁸ and feedback⁵⁸, and developing new metrics and measurements for human well-being^{44,90}.

Freedom and autonomy

Whereas some sources specifically refer to the freedom of expression^{28,73,82,105} or informational self-determination^{28,90} and ‘privacy-protecting user controls’⁵⁸, others generally promote freedom^{31,69,72}, empowerment^{28,52,99} or autonomy^{31,33,62,77,81,96}. Some documents refer to autonomy as a positive freedom, specifically the freedom to flourish³⁶, to self-determination through democratic means³⁸, the right to establish and develop relationships with other human beings^{38,92}, the freedom to withdraw consent⁶⁷, or the freedom to use a preferred platform or technology^{73,80}. Other documents focus on negative freedom, for example freedom from technological experimentation⁸², manipulation³³ or surveillance³⁸. Freedom and autonomy are believed to be promoted through transparency and predictable AI³⁸, by not ‘reducing options for and knowledge of citizens’³⁸, by actively increasing people’s knowledge about AI^{36,52,62}, giving notice and consent⁷⁹ or, conversely, by actively refraining from collecting and spreading data in absence of informed consent^{30,38,44,55,74}.

Trust

References to trust include calls for trustworthy AI research and technology^{50,97,99}, trustworthy AI developers and organisations^{51,60,66}, trustworthy ‘design principles’⁹¹, or underline the importance of customers’ trust^{23,52,58,66,74,80}. Calls for trust are proposed because a culture of trust among scientists and engineers is believed to support the achievement of other organisational goals⁹⁹, or because overall trust in the recommendations, judgments and uses of AI is indispensable for AI to ‘fulfill its world changing potential’²⁴. This last point is contradicted by one guideline explicitly warning against excessive trust in AI⁸¹. Suggestions for building or sustaining trust include education³³, reliability^{50,51}, accountability⁵⁶, processes to monitor and evaluate the integrity of AI systems over time⁵¹ and tools and techniques ensuring compliance with norms and standards^{43,63}. Whereas some guidelines require AI to be transparent^{37,43,57,58}, understandable^{36,37}, or explainable⁵² in order to build trust, another one explicitly suggests that, instead of demanding understandability, it should be ensured that AI fulfills public expectations⁵⁰. Other reported facilitators of trust include ‘a Certificate of Fairness’¹⁰⁴, multi-stakeholder dialogue⁶⁴, awareness about the value of using personal data⁷⁴, and avoiding harm^{30,56}.

Sustainability

To the extent that is referenced, sustainability calls for development and deployment of AI to consider protecting the environment^{33,38,46}, improving the planet’s ecosystem and biodiversity³⁷, contributing to fairer and more equal societies⁶⁵ and promoting peace⁶⁶. Ideally, AI creates sustainable systems^{44,76,90} that process data sustainably⁴³ and whose insights remain valid over time⁴⁸. To achieve this aim, AI should be designed, deployed and managed with care³⁸ to increase its energy efficiency and minimize its ecological footprint³¹. To make future developments sustainable, corporations are asked to create policies ensuring accountability in the domain of potential job losses³⁷ and to use challenges as an opportunity for innovation³⁸.

Dignity

While dignity remains undefined in existing guidelines, save the specification that it is a prerogative of humans but not robots⁹², there is frequent reference to what it entails: dignity is intertwined with human rights¹⁰¹ or otherwise means avoiding harm³¹, forced

acceptance³¹, automated classification³⁸, and unknown human-AI interaction³⁸. It is argued that AI should not diminish³³ or destroy⁸⁰ but respect⁸², preserve⁶⁹ or even increase human dignity^{36,37}. Dignity is believed to be preserved if it is respected by AI developers in the first place⁹⁶ and promoted through new legislation³⁸, through governance initiatives³⁶, or through government-issued technical and methodological guidelines⁸².

Solidarity

Solidarity is mostly referenced in relation to the implications of AI for the labor market¹⁰⁴. Sources call for a strong social safety net^{37,84}. They underline the need for redistributing the benefits of AI in order not to threaten social cohesion⁴⁹ and respecting potentially vulnerable persons and groups³³. Lastly, there is a warning of data collection and practices focused on individuals which may undermine solidarity in favour of ‘radical individualism’³⁸.

Discussion

We found a rapid increase in the number and variety of guidance documents for ethical AI, demonstrating the increasing active involvement of the international community. Organisations publishing AI guidelines come from a wide range of sectors. In particular the nearly equivalent proportion of documents issued by the public sector (i.e. governmental and inter-governmental organisations) and the private sector (companies and private sector alliances) indicate that the ethical challenges of AI concern both public entities and private enterprises. However, there is significant divergence in the solutions proposed to meet the ethical challenges of AI. Further, the relative underrepresentation of geographic areas such as Africa, South and Central America and Central Asia indicates that the international debate over ethical AI may not be happening globally in equal measures. MEDC countries are shaping this debate more than others, which raises concerns about neglecting local knowledge, cultural pluralism and global fairness.

The proliferation of soft-law efforts can be interpreted as a governance response to advanced research into AI, whose research output and market size have drastically increased¹⁰⁶ in recent years. Our analysis shows the emergence of an apparent cross-stakeholder convergence on promoting the ethical principles of transparency, justice, non-maleficence, responsibility, and privacy. Nonetheless, our thematic analysis reveals substantive divergences in relation to four major factors: (i) how ethical principles are interpreted, (ii)

why they are deemed important, (iii) what issue, domain or actors they pertain to, and (iv) how they should be implemented. Furthermore, unclarity remains as to which ethical principles should be prioritized, how conflicts between ethical principles should be resolved, who should enforce ethical oversight on AI and how researchers and institutions can comply with the resulting guidelines. These findings suggest the existence of a gap at the cross-section of principles formulation and their implementation into practice which can hardly be solved through technical expertise or top-down approaches.

Although no single ethical principle is explicitly endorsed by all existing guidelines, transparency, justice and fairness, non-maleficence, responsibility and privacy are each referenced in more than half of all guidelines. This focus could be indicating a developing convergence on ethical AI around these principles in the global policy landscape. In particular, the prevalence of calls for transparency, justice and fairness points to an emerging moral priority to require transparent processes throughout the entire AI continuum (from transparency in the development and design of algorithms to transparent practices for AI use), and to caution the global community against the risk that AI might increase inequality if justice and fairness considerations are not adequately addressed. Both these themes appear to be intertwined with the theme of responsibility, as the promotion of both transparency and justice seems to postulate increased responsibility and accountability on the side of AI makers and deployers.

It has been argued that transparency is not an ethical principle per se, but rather “a proethical condition for enabling or impairing other ethical practices or principles”¹⁰⁷. The proethical nature of transparency might partly explain its higher prevalence compared to other ethical principles. It is notable that current guidelines place significant value in the promotion of responsibility and accountability, yet few of them emphasize the duty of all stakeholders involved in the development and deployment of AI to act with integrity. This mismatch is probably associated with the observation that existing guidelines fail to establish a full correspondence between principles and actionable requirements, with several principles remaining uncharacterized or disconnected from the requirements necessary for their realization.

As codes related to non-maleficence outnumber those related to beneficence, it appears that, for the current AI community, the moral obligation to preventing harm takes precedence over the promotion of good. This fact can be partly interpreted as an instance of the so-called negativity bias, i.e. a general cognitive bias to give greater weight to negative entities^{108,109}. This negative characterization of ethical values is further emphasized by the fact that existing guidelines focus primarily on how to preserve privacy, dignity, autonomy and individual freedom *in spite of* advances in AI, while largely neglecting whether these principles could be promoted through responsible innovation in AI¹¹⁰.

The issue of trust in AI, while being addressed by less than one third of all sources, tackles a critical ethical dilemma in AI governance: determining whether it is morally desirable to foster public trust in AI. While several sources, especially those produced within the private sector, highlight the importance of fostering trust in AI through educational and awareness-raising activities, a smaller number of sources contend that trust in AI may actually diminish scrutiny and undermine some societal obligations of AI producers¹¹¹. This possibility would challenge the dominant view in AI ethics that building public trust in AI is a fundamental requirement for ethical governance¹¹².

The relative thematic underrepresentation of sustainability and solidarity suggests that these topics might be currently flying under the radar of the mainstream ethical discourse on AI. The underrepresentation of sustainability-related principles is particularly problematic in light of the fact that the deployment of AI requires massive computational resources which, in turn, require high energy consumption. The environmental impact of AI, however, does not only involve the negative effects of high-footprint digital infrastructures, but also the possibility of harnessing AI for the benefit of ecosystems and the entire biosphere. This latter point, highlighted in a report by the World Economic Forum¹¹³ though not in the AI guidelines by the same institution, requires wider endorsement to become entrenched in the ethical AI narrative. The ethical principle of solidarity is sparsely referenced, typically in association with the development of inclusive strategies for the prevention of job losses and unfair sharing of burdens. Little attention is devoted to promoting solidarity through the emerging possibility of using AI expertise for solving humanitarian challenges, a mission that is currently being pursued, among others, by intergovernmental organisations such as the United Nations Office for Project Services (UNOPS)¹¹⁴ or the World Health

Organization (WHO) and private companies such as Microsoft¹¹⁵. As the humanitarian cost of anthropogenic climate change is rapidly increasing¹¹⁶, the principles of sustainability and solidarity appear strictly intertwined though poorly represented compared to other principles.

While numerical data indicate an emerging convergence around the promotion of some ethical principles, in-depth thematic analysis paints a more complicated picture, as there are critical differences in *how* these principles are interpreted as well as what requirements are considered to be necessary for their realization. Results show that different and often conflicting measures are proposed for the practical achievement of ethical AI. For example, the need for ever larger, more diverse datasets to “unbias” AI appears difficult to conciliate with the requirement to give individuals increased control over their data and its use in order to respect their privacy and autonomy. Similar contrasts emerge between the requirement of avoiding harm at all costs and that of balancing risks and benefits. Furthermore, it should be noted that risk-benefit evaluations will lead to different results depending on whose well-being it will be optimized for by which actors. If not resolved, such divergences and tensions may undermine attempts to develop a global agenda for ethical AI.

Despite a general agreement that AI should be ethical, significant divergences emerge within and between guidelines for ethical AI. Furthermore, uncertainty remains regarding how ethical principles and guidelines should be implemented. These challenges have implications for science policy, technology governance and research ethics. At the policy level, they urge increased cooperative efforts among governmental organisations to harmonize and prioritize their AI agendas, an effort that can be mediated and facilitated by inter-governmental organisations. While harmonization is desirable, however, it should not come at the costs of obliterating cultural and moral pluralism over AI. Therefore, a fundamental challenge for developing a global agenda for AI is balancing the need for cross-national harmonization over the respect for cultural diversity and moral pluralism. This challenge will require the development of deliberative mechanisms to adjudicate disagreement concerning the values and implications of AI advances among different stakeholders from different global regions. At the level of technology governance, harmonization is typically implemented in terms of standardizations. Efforts in this direction have been made, among others, by the Institute of Electrical and Electronics Engineers

(IEEE) through the “Ethically Aligned Designed” initiative¹¹⁷. Finally, soft governance mechanisms such as Independent Review Boards (IRBs) will be increasingly required to assess the ethical validity of AI applications in scientific research, especially those in the academic domain. However, AI applications by governments or private corporations will unlikely fall under their oversight, unless significant expansions to the IRBs’ purview are made.

The international community seems to converge on the importance of transparency, non-maleficence, responsibility, and privacy for the development and deployment of ethical AI. However, enriching the current ethical AI discourse through a better appraisal of critical yet underrepresented ethical principles such as human dignity, solidarity and sustainability is likely to result into a better articulated ethical landscape for artificial intelligence. Furthermore, shifting the focus from principle-formulation to translation into practice must be the next step. A global agenda for ethical AI should balance the need for cross-national and cross-domain harmonization over the respect for cultural diversity and moral pluralism. Overall, our review provides a useful starting point for understanding the inherent diversity of current principles and guidelines for ethical AI and outlines the challenges ahead for the global community.

Limitations

This study has several limitations. First, guidelines and soft-law documents are an instance of gray literature, hence not indexed in conventional scholarly databases. Therefore, their retrieval is inevitably less replicable and unbiased compared to systematic database search of peer-reviewed literature. Following best practices for gray literature review, this limitation has been mitigated by developing a discovery and eligibility protocol which was pilot-tested prior to data collection. Although search results from search engines are personalized, the risk of personalization influencing discovery has been mitigated through the broadness of both the keyword search and the inclusion of results. A language bias may have skewed our corpus towards English results. Our content analysis presents the typical limitations of qualitative analytic methods. Following best practices for content analysis, this limitation has been mitigated by developing an inductive coding strategy which was conducted independently by two reviewers to minimize subjective bias. Finally, given the rapid pace of publication of AI guidance documents, there is a possibility that new policy

documents were published after our search was completed. To minimize this risk, continuous monitoring of the literature was conducted in parallel with the data analysis and until April 23, 2019.

Methods

We conducted a scoping review of the gray literature reporting principles and guidelines for ethical AI. A scoping review is a method aimed at synthesizing and mapping the existing literature¹¹⁸, which is considered particularly suitable for complex or heterogeneous areas of research^{118,119}. Given the absence of a unified database for AI-specific ethics guidelines, we developed a protocol for discovery and eligibility, adapted from the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) framework¹²⁰. The protocol was pilot-tested and calibrated prior to data collection. Following best practices for gray literature retrieval, a multi-stage screening strategy involving both inductive screening via search engine and deductive identification of relevant entities with associated websites and online collections was conducted. To achieve comprehensiveness and systematicity, relevant documents were retrieved by relying on three sequential search strategies (cf. Figure 2): First, a manual search of four link hub webpages (“linkhubs”)^{121–124} was performed. 68 sources were retrieved, out of which 30 were eligible (27 after removing duplicates). Second, a keyword-based web search of the Google.com search engine was performed in private browsing modus, after log-out from personal accounts and erasure of all web cookies and history.^{125,126} Search was performed using the following keywords: [AI principles], [artificial intelligence principles], [AI guidelines], [artificial intelligence guidelines], [ethical AI] and [ethical artificial intelligence]. Every link in the first thirty search results was followed and screened (i) for AI principles, resulting in 10 more sources after removing duplicates, and (ii) for articles mentioning AI principles, leading to the identification of 3 additional non-duplicate sources. The remaining Google results up to the 200th listings for each Google search were followed and screened for AI principles only. Within these additional 1020 link listings we identified 15 non-duplicate documents. After identifying relevant documents through the two processes above, we used citation-chaining to manually screen the full-texts and, if applicable, reference lists of all eligible sources in order to identify other relevant documents. 17 additional sources were identified. We continued to monitor the literature in parallel with the data analysis and until April 23, 2019, to retrieve eligible documents that were released after our search was completed. Twelve

English, German, French, Italian, Greek; (ii) issued by institutional entities from both the public and the public sectors; (iii) referred explicitly in their title/description to AI or ancillary notions, (iv) expressed a normative ethical stance defined as a moral preference for a defined course of action (cf. SI Table S2). Following full-text screening, 84 sources or parts thereof were included in the final synthesis (cf. SI Table S1).

Content analysis of the 84 sources was independently conducted by two researchers in two cycles of manual coding and one cycle of code mapping within the qualitative data analysis software *Nvivo* for Mac v.11.4. During the first cycle of coding, one researcher exhaustively tagged all relevant text through inductive coding¹²⁸ attributing a total of 3457 codes, out of which 1180 were subsequently discovered to pertain to ethical principles. Subsequently, two researchers conducted the code mapping process in order to reduce subjective bias. The process of code mapping, a method for qualitative metasynthesis¹²⁹, consisted of two iterations of themeing¹²⁸, whereby categories were first attributed to each code, then categorized in turn (cf. SI Table S3). For the theming of ethical principles, we relied deductively on normative ethical literature. Ethical categories were inspected and assessed for consistency by two researchers with primary expertise in ethics. Thirteen ethical categories emerging from code mapping, two of which were merged with others due to independently assessed semantic and thematic proximity. Finally, we extracted significance and frequency by applying focused coding, a second cycle coding methodology used for interpretive analysis¹²⁸, to the data categorized in ethical categories. Consistency check was performed both by reference to the relevant ethical literature and a process of deliberative mutual adjustment among the general principles and the particular judgments contained in the policy documents, an analytic strategy known as ‘reflective equilibrium’¹³⁰.

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Supplementary Information for

Artificial Intelligence: the global landscape of ethics guidelines

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Tables S1 to S3

Table S1. Ethics guidelines for AI by date of publishing (incl. details)

Name of Document/Website	Name of guidelines/principles	Issuer	Country of issuer	Type of issuer	Date of publishing	Target audience	Retrieval
Principles of robotics	Principles for designers, builders and users of robots	Engineering and Physical Sciences Research Council UK (EPSRC)	UK	Science foundation	1-Apr-2011	multiple (public, developers)	Linkhubs
Ethique de la recherche en robotique	Préconisations	CERNA (Allistene)	France	Research alliance	xx-Nov-2014	researchers	Citation chaining
Unified Ethical Frame for Big Data Analysis. IAF Big Data Ethics Initiative, Part A	Values for an Ethical Frame	The Information Accountability Foundation	UK	NPO/Charity	xx-Mar-2015	unspecified	Citation chaining
Ethics Policy	IIIM's Ethics Policy	Icelandic Institute for Intelligent Machines (IIIM)	Iceland	Academic and research institution	31-Aug-2015	self	Linkhubs
The AI Now Report. The Social and Economic Implications of Artificial Intelligence Technologies in the Near-Term	Key recommendations	AI Now Institute	USA	Academic and research institution	22-Sep-2016	unspecified	Citation chaining
Tenets	Tenets	Partnership on AI	n.a.	Private sector alliance	29-Sep-2016	self	Web search results 1-30
Preparing for the future of Artificial Intelligence	Recommendations in this Report	Executive Office of the President; National Science and Technology Council; Committee on Technology	USA	Governmental agencies/organizations	xx-Oct-2016	multiple (stakeholders engaged at various points in the production, use, governance, and assessment of AI systems)	Linkhubs
The National Artificial Intelligence Research and Development Strategic Plan	R&D Strategy	National Science and Technology Council; Networking and Information Technology Research and Development Subcommittee	USA	Governmental agencies/organizations	xx-Oct-2016	self	Linkhubs
Position on Robotics and Artificial Intelligence	3. Principles // 6. Recommendations Green position on Robotics and Artificial Intelligence	The Greens (Green Working Group Robots)	EU	Political Party	22-Nov-2016	multiple (EU parliament, public, self)	Web search results 31-200
Principles for Accountable Algorithms and a Social Impact Statement for Algorithms	Principles for Accountable Algorithms	Fairness, Accountability, and Transparency in Machine Learning (FATML)	n.a.	n.a.	24-Nov-2016	multiple (developers and product managers)	Linkhubs
Statement on Algorithmic Transparency and Accountability	Principles for Algorithmic Transparency and Accountability	Association for Computing Machinery (ACM)	USA	Prof. Association/Society	12-Jan-2017	multiple (developers, deployers)	Linkhubs
Report with recommendations to the Commission on Civil Law Rules on Robotics	Motion for a European Parliament Resolution	European Parliament	EU	IGO/supra-national	27-Jan-2017	public sector (lawmakers)	Linkhubs
AI Principles	AI Principles	Future of Life Institute	USA	Miscellaneous (mixed crowdsourced, NPO)	30-Jan-2017	unspecified	Linkhubs
The Japanese Society for Artificial Intelligence Ethical Guidelines	The Japanese Society for Artificial Intelligence Ethical Guidelines	Japanese Society for Artificial Intelligence	Japan	Prof. Association/Society	28-Feb-2017	self (incl AI)	Linkhubs
Report on Artificial Intelligence and Human Society (Unofficial translation)	4.1 Ethical issues	Advisory Board on Artificial Intelligence and Human Society (initiative of the Minister of State for Science and Technology Policy)	Japan	Governmental agencies/organizations	24-Mar-2017	multiple (researchers, government, businesses, public, educators)	Web search results 31-200
Artificial Intelligence and Machine Learning: Policy Paper	Guiding principles and recommendations	Internet Society	international	NPO/charity	18-Apr-2017	multiple (policy-makers, other stakeholders in the wider Internet ecosystem)	Web search results 31-200
Machine learning: the power and promise of computers that learn by example	Chapter six – A new wave of machine learning research	The Royal Society	UK	Prof. Association/Society	xx-Apr-2017	unspecified	Citation chaining
The Ethics of Code: Developing AI for Business with Five Core Principles	The Ethics of Code: Developing AI for Business with Five Core Principles	Sage	UK	Company	27-Jun-2017	self	Citation chaining
Automated and Connected Driving: Report	Ethical rules for automated and connected vehicular traffic	Federal Ministry of Transport and Digital Infrastructure, Ethics Commission	Germany	Governmental agencies/organizations	xx-Jun-2017	multiple (automated & connected vehicular traffic)	Linkhubs

Mid- to Long-Term Master Plan in Preparation for the Intelligent Information Society	Tasks (8-12)	Government of the Republic of Korea	South Korea	Governmental agencies/organizations	20-Jul-2017	self (gov)	Linkhubs
Draft AI R&D Guidelines for International Discussions	AI R&D Principles	Institute for Information and Communications Policy (IICP), The Conference toward AI Network Society	Japan	Governmental agencies/organizations	28-Jul-2017	multiple (systems and developers)	Linkhubs
Big data, artificial intelligence, machine learning and data protection	Key recommendations	Information Commissioner's Office	UK	Gov	4-Sep-2017	organisations	Web search results 1-30
Report of COMEST on Robotics Ethics (only section "Recommendations" taken into account)	Relevant ethical principles and values	COMEST/UNESCO	international	IGO/supra-national	14-Sep-2017	unspecified	Citation chaining
Ethical Principles for Artificial Intelligence and Data Analytics	Ethical Principles for Artificial Intelligence and Data Analytics	Software & Information Industry Association (SIIA), Public Policy Division	international	Private sector alliance	15-Sep-2017	private sector (industry organizations)	
AI - Our approach	AI - Our approach	Microsoft	USA	Company	7-Oct-2017	self	Web search results 1-30
DeepMind Ethics & Society Principles	Our Five Core Principles	DeepMind Ethics & Society	UK	Company	10-Oct-2017	self	Citation chaining
Human Rights in the Robot Age Report	Recommendations	The Rathenau Institute	Netherlands	Academic and research institution (Gov)	11-Oct-2017	public sector (Council of Europe)	Citation chaining
Artificial Intelligence. The Public Policy Opportunity	Summary of Recommendations	Intel Corporation	USA	Company	18-Oct-2017	public sector (policy makers)	Citation chaining
ITI AI Policy Principles	ITI AI Policy Principles	Information Technology Industry Council (ITI)	international	Private sector alliance	24-Oct-2017	self (members)	Citation chaining
AI Now 2017 Report	Recommendations, Executive Summary	AI Now Institute	USA	Academic and research institution	xx-Oct-2017	multiple (core public agencies, companies, industry, universities, conferences, other stakeholders)	Citation chaining
Montréal Declaration: Responsible AI	Montréal Declaration: Responsible AI	Université de Montréal	Canada	Academic and research institution	3-Nov-2017	multiple (public, developers, policy makers)	Linkhubs
Ethically Aligned Design. A Vision for Prioritizing Human Well-being with Autonomous and Intelligent Systems, version 2	Ethically Aligned Design. A Vision for Prioritizing Human Well-being with Autonomous and Intelligent Systems, version 2	Institute of Electrical and Electronics Engineers (IEEE), The IEEE Global Initiative on Ethics of Autonomous and Intelligent Systems	international	Prof. Association/Society	12-Dec-2017	unspecified	Linkhubs
How can humans keep the upper hand? Report on the ethical matters raised by AI algorithms (only section "From principles to policy recommendations")	From principles to policy recommendations	French Data Protection Authority (CNIL)	France	Governmental agencies/organizations	15-Dec-2017	unspecified	Linkhubs
Top 10 Principles for Ethical Artificial Intelligence	Top 10 Principles for Ethical Artificial Intelligence	UNI Global Union	international	Federation/Union	17-Dec-2017	multiple (unions, workers)	Linkhubs
Business Ethics and Artificial Intelligence	Fundamental Values and Principles	Institute of Business Ethics	UK	Private sector alliance	11-Jan-2018	private sector (users of AI in business)	Web search results 31-200
IBM's Principles for Trust and Transparency	IBM's Principles for Trust and Transparency	IBM	USA	Company	17-Jan-2018	self	Web search results 1-30
Artificial intelligence and privacy	Recommendations for privacy friendly development and use of AI	The Norwegian Data Protection Authority	Norway	Governmental agencies/organizations	xx-Jan-2018	multiple (developers, system suppliers, organizations, end users, authorities)	Web search results 31-200
The Malicious Use of Artificial Intelligence: Forecasting, Prevention, and Mitigation	Four High-Level Recommendations	Future of Humanity Institute; University of Oxford; Centre for the Study of Existential Risk; University of Cambridge; Center for a New American Security; Electronic Frontier Foundation; OpenAI	international	Miscellaneous (mixed academic, NPO)	20-Feb-2018	unspecified	Citation chaining
White Paper: How to Prevent Discriminatory Outcomes in Machine Learning	Executive summary	WEF, Global Future Council on Human Rights 2016-2018	international	NPO/Charity	12-Mar-2018	private sector (companies)	Citation chaining

For a meaningful Artificial Intelligence. Towards a French and European strategy	"Part 5 — What are the Ethics of AI?; Part 6 — For Inclusive and Diverse Artificial Intelligence"	Mission Villani	France	Governmental agencies/organizations	29-Mar-2018	public sector (French government/parliament)	Linkhubs
Statement on Artificial Intelligence, Robotics and 'Autonomous' Systems	Ethical principles and democratic prerequisites	European Commission, European Group on Ethics in Science and New Technologies	EU	IGO/supra-national	xx-Mar-2018	public sector (EU Commission)	Linkhubs
L'intelligenza artificiale al servizio del cittadino	Sfida 1: Etica	Agenzia per l'Italia Digitale (AGID)	Italy	Governmental agencies/organizations	xx-Mar-2018	multiple (government, schools, healthcare institutions)	Linkhubs
OpenAI Charter	OpenAI Charter	OpenAI	USA	NPO/charity(*)	9-Apr-2018	self	Linkhubs
AI in the UK: ready, willing and able? (report, only section "An AI Code" taken into account)	no title. P. 125: "... we suggest five overarching principles for an AI Code."	UK House of Lords, Select Committee on Artificial Intelligence	UK	Governmental agencies/organizations	16-Apr-2018	public sector (UK government)	Linkhubs
Privacy and Freedom of Expression In the Age of Artificial Intelligence	Conclusions and Recommendations	Privacy International & Article 19	international	NPO/Charity	25-Apr-2018	multiple (states, companies, civil society)	Citation chaining
AI Guidelines	AI Guidelines	Deutsche Telekom	Germany	Company	11-May-2018	self	Web search results 1-30
The Toronto Declaration: Protecting the right to equality and non-discrimination in machine learning systems	The Toronto Declaration: Protecting the right to equality and non-discrimination in machine learning systems	Access Now ; Amnesty International	international	Miscellaneous (mixed NGO, NPO)	16-May-2018	multiple (states, private sector actors)	Linkhubs
Discussion Paper on Artificial Intelligence (AI) and Personal Data - Fostering Responsible Development and Adoption of AI	Principles for responsible AI	Personal Data Protection Commission Singapore	Singapore	Governmental agencies/organizations	5-Jun-2018	multiple (business; Trade associations and chambers, professional bodies and interest groups)	Linkhubs
Our principles	Our principles	Google	USA	Company	7-Jun-2018	self	Web search results 1-30
Discussion Paper: National Strategy for Artificial Intelligence (only section "Ethics, Privacy, Security and Artificial Intelligence. Towards a "Responsible AI"")	Ethics, Privacy, Security and Artificial Intelligence. Towards a "Responsible AI"	National Institution for Transforming India (Niti Aayog)	India	Governmental agencies/organizations	8-Jun-2018	self (Indian government)	Linkhubs
Charlevoix Common Vision for the Future of Artificial Intelligence	Charlevoix Common Vision for the Future of Artificial Intelligence	Leaders of the G7	international	IGO/supra-national	9-Jun-2018	self (gov)	Linkhubs
Policy Recommendations on Augmented Intelligence in Health Care H-480.940	Policy Recommendations on Augmented Intelligence in Health Care H-480.940	American Medical Association (AMA)	USA	Prof. Association/Society	14-Jun-2018	self	Web search results 31-200
Artificial Intelligence: open questions about gender inclusion	Proposals	W20	international	IGO/supra-national	2-Jul-2018	public sector (states/countries)	Web search results 31-200
Everyday Ethics for Artificial Intelligence. A practical guide for designers & developers	Five Areas of Ethical Focus	IBM	USA	Company	2-Sep-2018	designers	Web search results 1-30
Artificial Intelligence (AI) in Health	Key recommendations	Royal College of Physicians	UK	Prof. Association/Society	3-Sep-2018	multiple (industry, doctors, regulators)	Web search results 31-200
Initial code of conduct for data-driven health and care technology	10 Principles	UK Department of Health & Social Care	UK	Governmental agencies/organizations	5-Sep-2018	developers	Web search results 31-200
Work in the age of artificial intelligence. Four perspectives on the economy, employment, skills and ethics (only section "Good application of artificial intelligence technology and ethics")	Values of a good artificial intelligence society	Ministry of Economic Affairs and Employment	Finland	Governmental agencies/organizations	10-Sep-2018	multiple (Finnish world of work)	Linkhubs
SAP's guiding principles for artificial intelligence	SAP's guiding principles for artificial intelligence	SAP	Germany	Company	18-Sep-2018	self	Web search results 1-30
Sony Group AI Ethics Guidelines	Sony Group AI Ethics Guidelines	SONY	Japan	Company	25-Sep-2018	self (group)	Web search results 1-30
Ethics Framework - Responsible AI	Framework	Machine Intelligence Garage Ethics Committee	UK	n.a.	28-Sep-2018	private sector (start-ups)	Web search results 31-200

Dutch Artificial Intelligence Manifesto	Multidisciplinary challenges	Special Interest Group on Artificial Intelligence (SIGAI), ICT Platform Netherlands (IPN)	Netherlands	Academic and research institution	xx-Sep-2018	multiple (Dutch government, researchers)	Web search results 31-200
Governing Artificial Intelligence. Upholding Human Rights & Dignity	Recommendations	Data & Society	USA	Research (NPO)	10-Oct-2018	multiple (companies, researchers, governments, policy makers, UN)	Citation chaining
Tieto's AI ethics guidelines	Tieto's AI ethics guidelines	Tieto	Finland	Company	17-Oct-2018	self	Web search results 31-200
Intel's AI Privacy Policy White Paper. Protecting individuals' privacy and data in the artificial intelligence world	Six Policy Recommendations	Intel Corporation	USA	Company	22-Oct-2018	public sector (policy makers)	Web search results 31-200
Universal Guidelines for Artificial Intelligence	Universal Guidelines for Artificial Intelligence	The Public Voice	international	Mixed (coalition of NGOs, ICOs etc.)	23-Oct-2018	multiple (institutions, governments)	Web search results 1-30
Declaration on ethics and data protection in Artificial Intelligence	"... guiding principles ..."	ICDPPC	international	IGO/supra-national	23-Oct-2018	unspecified	Web search results 1-30
AI Principles of Telefónica	AI Principles of Telefónica	Telefonica	Spain	Company	30-Oct-2018	self	Web search results 1-30
Introducing Unity's Guiding Principles for Ethical AI – Unity Blog	Unity's six guiding AI principles are as follows	Unity Technologies	USA	Company	28-Nov-2018	self	Manual inclusion
Responsible bots: 10 guidelines for developers of conversational AI	Guideline	Microsoft	USA	Company	xx-Nov-2018	developers	Manual inclusion
AI Now 2018 Report	Recommendations	AI Now Institute	USA	Academic and research institution	xx-Dec-2018	multiple	Manual inclusion
Ethics of AI in Radiology: European and North American Multisociety Statement	Conclusion	American College of Radiology; European Society of Radiology; Radiology Society of North America; Society for Imaging Informatics in Medicine; European Society of Medical Imaging Informatics; Canadian Association of Radiologists; American Association of Physicists in Medicine	international	Prof. Association/Society	26-Feb-2019	self	Manual inclusion
European ethical Charter on the use of Artificial Intelligence in judicial systems and their environment	"The five principles of the Ethical Charter on the Use of Artificial Intelligence in Judicial Systems and their environment"	Concil of Europe: European Commission for the efficiency of Justice (CEPEJ)	EU	IGO/supra-national	xx-Feb-2019	multiple (public and private stakeholders)	Manual inclusion
Ethically Aligned Design: A Vision for Prioritizing Human Well-being with Autonomous and Intelligent Systems, First Edition (EAD1e)	General Principles	Institute of Electrical and Electronics Engineers (IEEE), The IEEE Global Initiative on Ethics of Autonomous and Intelligent Systems	international	Prof. Association/Society	25-Mar-2019	multiple (technologists, educators, and policy maker)	Manual inclusion
Artificial Intelligence. Australia's Ethics Framework. A discussion Paper	Core principles for AI; A toolkit for ethical AI	Department of Industry Innovation and Science	Australia	Governmental agencies/organizations	5-Apr-2019	unspecified	Manual inclusion
Ethics Guidelines for Trustworthy AI	Ethical Principles in the Context of AI Systems	High-Level Expert Group on Artificial Intelligence	EU	IGO/supra-national	8-Apr-2019	multiple (all stakeholders)	Manual inclusion
Ethical, social, and political challenges of Artificial Intelligence in Health	Conclusion	Future Advocacy	UK	Company	xx-Apr-2019	unspecified	Manual inclusion
The responsible AI framework	Operating AI	PriceWaterhouse-Coopers UK	UK	Company	n.a.	multiple (clients)	Web search results 31-200
Digital Decisions	VI. Solutions Part 1: Principles	Center for Democracy & Technology	USA	NPO/charity	n.a.	unspecified	Citation chaining
Responsible AI and robotics. An ethical framework	Our view	Accenture UK	UK	Company	n.a.	private sector	Web search results 1-30
Commitments and principles	OP Financial Group's ethical guidelines for artificial intelligence	OP Group	Finland	Company	n.a.	self	Web search results 31-200
Science, Law and Society (SLS) Initiative	Principles for the Governance of AI	The Future Society	USA	NPO/charity	n.a.	public sector (policy makers)	Linkhubs

10 Principles of responsible AI	Summary of our proposed Recommendations	Women leading in AI	n.a.	n.a.	n.a.	public sector (national and international policy makers)	Manual inclusion
AI4People—An Ethical Framework for a Good AI Society: Opportunities, Risks, Principles, and Recommendations	Action Points	AI4People	EU	n.a.	n.a.	unspecified	Manual inclusion
AI Principles & Ethics	AI Principles; AI guidelines	Smart Dubai	UAE	Governmental agencies/organizations	n.a. 2018?	self	Manual inclusion

Table S2. Screening and Eligibility (details)

Screening	
Sources considered:	- Types: websites and documents published online or parts thereof such as policy documents, principles, guidelines, recommendations, dedicated webpages, institutional reports and declarations; - Issuers: institutions, associations and organizations such as companies, corporations, NGOs, NPOs, academic and professional societies, governmental institutions and affiliated organizations; - Language: English, German, French, Italian, Greek (the languages spoken by the researchers).
Sources excluded:	- Types: videos, images, audio/podcasts, books, blog articles, academic articles, journalistic articles, syllabi, legislation, official standards, conference summaries; - Issuers: individual authors; - Language: others than those above.
Eligibility	
Sources included:	- which refer to “artificial intelligence” and/or “AI”, either explicitly in their title or within their description (example: UK, House of Lords: “AI in the UK: ready, willing and able”); or - which do not contain the above reference in their title but mention “robot” or “robotics” instead <i>and</i> reference AI or artificial intelligence explicitly as being part of robots and/or robotics (example: “Principles of robotics”); or - which do not contain the above reference in their title but are thematically equivalent (by referring to “algorithms”, “predictive analytics”, “cognitive computing”, “machine learning”, “deep learning”, “autonomous” or “automated” instead (example: “Automated and Connected Driving: Report”). - which self-proclaim to be a principle or guideline (including “ethics/ethical”, “principles”, “tenets”, “declaration”, “policy”, “guidelines”, “values” etc.); or - which is expressed in normative or prescriptive language (i.e. with modal verbs or imperatives); or - which is principle- or value-based (i.e. indicating a preference and/or a commitment to a certain ethical vision or course of action).
Excluded sources:	- websites and documents about robotics that do not mention artificial intelligence as being part of robots/robotics; and - websites and documents about data or data ethics that do not mention artificial intelligence as being part of data;

Table S3. Categorization after themeing and code mapping

Question addressed	Thematic family	Themes
What?	Ethical Principles & Values	Ethical Principles I. Beneficence II. Non-maleficence III. Trust IV. Transparency & Explainability V. Freedom and autonomy (incl. consent) VI. Privacy VII. Justice, Fairness & Equity VIII. Responsibility & Accountability IX. Dignity X. Sustainability XI. Solidarity
		Specific functionalities I. Feedback & feedback-loop II. Decision-making
	Technical and methodological aspects	Data & datasets I. Data origin/input II. Data use III. Metadata IV. Algorithms
		Methodological challenges I. Methodology II. Metrics & measurements III. Tests, testing IV. Ambiguity & uncertainty V. Accuracy VI. Reliability VII. Evidence and validation VIII. Black-box (opacity) IX. Data security X. Quality (of data/system/etc.)
		Benefits I. AI strengths, advantages II. Knowledge III. Innovation IV. Enhancement
Who?	Impact	Risks I. Risks II. Malfunction III. Misuse & dual-use IV. Deception V. Discrimination (duplicate in Justice&Fairness) VI. Surveillance VII. Manipulation VIII. Arms race
		Impact assessment I. Impact II. Goals/Purposes/Intentions III. Public opinion IV. Risk evaluation & mitigation (duplicate in Risks) V. Monitoring/Precaution VI. Future of work
Who?	Design & development	I. Industry II. AI researchers III. Designers IV. Developers
	Users	I. End users II. Organisations III. Public sector actors IV. Military V. Communities

	Specific stakeholders	I. Ethical and/or auditing committees II. Government III. Policy makers IV. Researchers & scientists V. Vulnerable groups & minorities
How?	Social engagement	I. Knowledge commons II. Education & training III. Public deliberation & democratic processes IV. Stakeholder involvement & partnerships
	Soft policy	I. Standards II. Certification III. Best practices IV. Whistleblowing
	Economic incentives	V. Business model & strategy VI. Funding & investments VII. Taxes/taxation
	Regulation & audits	VIII. Laws & regulation (general) IX. Data protection regulation X. IP law XI. Human rights treaties XII. Other rights & laws XIII. Audits & auditing